



PAK-MED STUDY CORNER

Your Success, Our Mission

**TOTAL MCQS**
180**MOCK TEST – 180 MCQS****TEST 4****BIO****CHE****PHY****ENG****LOGICAL****BIOLOGY**

- 01** Which one of the following is first line of defense?
A Macrophages B neutrophils
C natural killer cells D hydrochloric acid
- 02** Antigens may be
A Virus B bacteria
C fungi D all of these
- 03** Which one of the following is produced by immune system?
A Antigen B antibody
C antigen and antibody D none of these
- 04** Study of our protection from macromolecules or invading organism and our response for them
A Immunology B immune system
C protection D none of them
- 05** Monocytes flow in the blood stream when enter to tissues it turn into _ _ _
A Neutrophils B macrophages
C chemical messengers D none of these
- 06** The most common form of WBCs in the body are
A Macrophages B neutrophils
C Nk cells D none of these
- 07** Natural killer cells are type of lymphocytes and a component of _ _ _
A Innate immunity B adaptive immunity
C both A&B D digestive system
- 08** Natural killer cells play a vital role in host rejection of _ _ _
A Tumors B virally infected cells
C tumors and virally infected cells D liver
- 09** NK cells are cytotoxic, having small granules in cytoplasm in the form of proteins such as-----
A perforin B protease
C granzyme D all of these
- 10** The cell in the blood that detect the pathogenic bacteria and signal the complement system to get to work are
A Macrophages B antibodies
C natural killer cells D neutrophils
- 11** Major histocompatibility complex class II which display foreign peptides for
A Helper T cells B killer T cells
C macrophages D suppressor T cells
- 12** The chemicals from injured tissues cause blood vessels to leak fluid into tissue causing
A Disease B inflammation/swelling
C allergy D all of these
- 13** Which one of the following is short term immunity?
A Innate immunity B natural immunity
C active immunity D passive immunity
- 14** Vaccine may be prepared from
A Lived modified organisms
B inactivated or killed organisms
C Extracted cellular fractions toxoids
D combination of these
- 15** cell mediated immune response consists of
A T-cells B B-cells
C B-cells and T-cells D none of these
- 16** Antibodies are chemically proteins found in the
A Blood plasma B lymph
C blood plasma and lymph D hormones
- 17** Each antibody consists of four polypeptide chains
A 2 light 2 heavy B 1 light 2 heavy
C 3 light 1 heavy D 4 heavy
- 18** Active immunity is developed by _ _ _ ways
A By disease B by clinical infection
C by killed micro-organisms D all these

Quick Tips

- ✓ Read the question carefully
- ✓ Eliminate wrong options
- ✓ Manage time smartly
- ✓ Believe in yourself ✨



**“Discipline Today
Doctor Tomorrow!”**

Stay Focused • Keep Practicing • Never Give Up

**Test Info**

- ✓ Total MCQs: 180
- ✓ Subjects: 5
- ✓ This: Biology
- ✓ Test No: 4



19 Antibodies in mother's breast milk provide a baby with temporary immunity which is

- A** Innate immunity
- B** natural immunity
- C** active immunity
- D** passive immunity

20 Which one of the following is second line of defense?

- A** Skin
- B** mucus
- C** cilia
- D** macrophages

21 Skin provides * _ * _ * _

- A** Insulation
- B** temperature regulation
- C** sensation
- D** protection
- E** all these

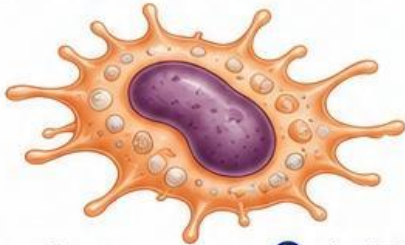
22 Epidermal dendritic cells that actively patrol the skin to phagocytize pathogens are present in

- A** Epidermis
- B** dermis
- C** hypodermis
- D** none of these

23 Which one of the following directly kills invaders?

- A** Cytotoxic T cells
- B** helper T cells
- C** suppressor cells
- D** all of these

24 Identify the type of WBC in following image



- A** Macrophage
- B** dendritic cell
- C** NK cell
- D** neutrophil

25 Which of the following are/is antigen presenting cell/cells (APCs)?

- A** Macrophages
- B** Dendritic cells
- C** NK cells
- D** Both A & B

26 Greater the concentration of water molecules, greater will be the kinetic energy of water, this is * _ * _

- A** Water potential
- B** osmotic pressure
- C** solute potential
- D** pressure potential

27 The change in water potential due to the presence of the solute molecules, this is called

- A** Osmotic pressure
- C** Osmotic potential/solute potential
- D** water potential

28 Osmotic potential is equal to * _ * _

- A** Water potential
- B** solute potential
- C** osmotic pressure
- D** pressure potential

29 The turgor pressure produced by endosmosis of cell sap is called

- A** Water potential
- B** solute potential
- C** osmotic pressure
- D** pressure potential

30 Water potential and solute potential is measured in

- A** pascal
- B** kilopascal
- C** mm
- D** kw

31 Solute potential + pressure potential =

- A** Water potential
- B** osmotic potential
- C** osmotic pressure
- D** all of these

32 Pure water has

- A** Zero water potential
- C** very low water potential
- C** Minimum water potential
- D** maximum water potential



33 Sieve tubes are specially adopted for the process of * _ * _

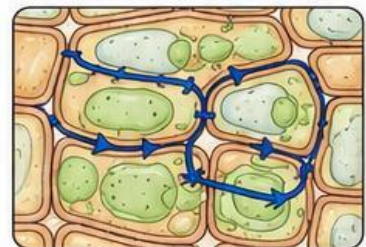
- A** Adhesion
- B** cohesion
- C** transformation
- D** translocation

34 Vessel elements are individual cells linked together end to end to form

- A** Xylem vessels
- B** phloem vessels
- C** both A and B
- D** none of these

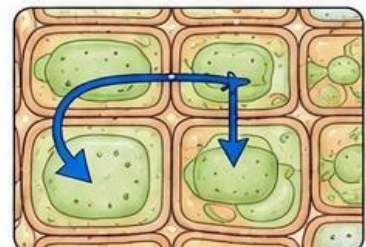
35 The whole nonliving pathway of water is called * _ * _

- A** apoplast pathway
- C** symplast pathway
- C** vacular pathway
- D** all of these



36 A living transport of water through cortical cells by osmosis is called

- A** apoplast pathway
- C** symplast pathway
- C** vacular pathway
- D** all of these



37 The water which enters to the epidermal cells of root passes through cortex, endodermis, pericycle and finally to the _____

- A** Xylem cells
- B** phloem cells
- C** both of these
- D** none of these

38 Identify the insectivorous/carnivorous plant in following image



- A** Sundew
- B** venus fly trap
- C** water fly trap
- D** pitcher plant

39 The science of Genetics originated in the year 1900 with the rediscovery of an article originally Published in 1865 by _____ named Gregor Johan mendel

- A** Teacher
- B** professor
- C** scientist
- D** monk

40 Mendel crossed $RrYy \times RrYy$ and the round green in result was _____

- A** 9/16
- B** 3/16
- C** 1/16
- D** 0/16

41 When neither of the two alleles express independently in heterozygous state is called

- A** Complete dominance
- B** incomplete dominance
- C** Co-dominance
- D** over dominance

42 Donor for blood group type AB is _____

- A** A, O
- B** B, O
- C** A, B, AB, O
- D** O

43 Which one of the following is not main character of pea plant?

- A** Short life cycle
- B** self pollinating flowers
- C** Cross pollination is possible
- D** need a large area to grow

44 Mendel observed that the recessive character appeared in the F₂ offspring in an average ratio of

- A** 3:1
- B** 1:3
- C** 4:1
- D** 1:1

45 Which one of the following law is not mendel's law of heredity?

- A** Law of dominance
- B** law of segregation
- C** Law of independent assortment
- D** none of these

46 A chance that a man will inherit the colour blind trait from a carrier mother is _____

- A** 10%
- B** 50%
- C** 90%
- D** 100%

47 Which one of the following is x linked recessive disorders?

- A** haemophilia
- B** colour blindness
- C** Diabetes insipidus
- D** all of these

48 Heterozygous individual (F₁) is crossed with recessive parent (P₁) if more parental and less Recombinant are produced than it is _____

- A** No linkage
- B** incomplete/partial linkage
- C** complete linkage
- D** none of these

49 Which is not related to colour blindness?

- A** Zigzag pattern of inheritance
- B** passes directly from father to son
- C** Opsins
- D** more common in male

50 ABO blood grouping is controlled by gene I which has three alleles and shows co-dominance. There are six genotypes. How many phenotypes in all are possible?

- A** Six
- B** three
- C** four
- D** five

51 A man whose father was hemophilic marries a woman who had a hemophilic mother and Normal father. What percentage of male children of this couple will be hemophilic?

- A** 50% **B** 25%
C 75% **D** 0%

52 In his classic experiments on pea plants, Mendel did not use

- A** Pod length **B** seed shape
C height of plant **D** position of flowers

53 Which term describes the alternative forms of a factor controlling a particular characteristic?

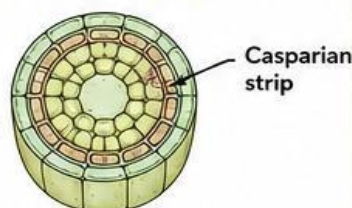
- A** Alleles **B** chromosomes
C genes **D** genotypes

54 In peas, the allele for tall plants, T, is dominant to the allele for dwarf plants, t. Which alleles would be present in the male nucleus of a pollen grain produced by a dwarf plant?

- A** T **B** t
C TT **D** tt

55 The Casparian strips are present in * _ *

- A** Cortex cells of root
B pericycle
C endodermal cells of root
D xylem



56 MN blood group system is an example of * _ *

- A** Complete dominance **B** incomplete dominance
C co-dominance **D** over-dominance

57 International society of blood transfusion has found more than * _ * types of blood group system

- A** 13 **B** 23
C 4 **D** 30

58 Crossing over in diploid organism is responsible for

- A** Dominance of genes
B linkage between genes
C Segregation of alleles
D recombination of linked alleles

59 How many scientists rediscovered Mendelism in 1900 independently?

- A** 1 **B** 2
C 3 **D** 4

60 In the birds-butterflies and moth flies, which type of sex determination is found?

- A** XO **B** XY type
C ZW type **D** any of these

61 Rh factor may be responsible for

- A** Turner's syndrome **B** AIDS
C sickle cell anaemia **D** Erythroblastosis fetalis

62 How many genotype will be produced by crossing of two alleles "A" and "a"?

- A** One **B** two
C three **D** four

63 If two heterozygous tall plants are crossed together, the proportion of phenotypically Tall plants will be

- A** 50%
C 25%
C 75%
D 100%

	T	t
T	TT	Tt
t	Tt	tt

64 Law of independent assortment can not be applied on

- A** Dominant genes **B** recessive genes
C linked genes **D** autosomal genes

65 A woman is homozygous for A-ve(negative) blood type. A man has AB-ve blood type. What is The probability that couple's child will be type B-ve(negative)

- A** 0% **B** 25%
C 50% **D** 75% **E** 100%

66 AB blood group type contain antibodies for

- A** B **B** A **C** A, B **D** none

67 T.H Morgan when crossed normal grey with vestigial black, the number of produced grey normal was in test cross

- A** 965 **B** 944 **C** 216
C 185 **D** 185

68 The flower position which is dominant is * _ *



Axial
(position of flowers)



Terminal
(position of flower at tip)

- A** Axial **B** terminal **C** mid **D** none of these

69 The electronic configuration of an element is $2\ 8/1$ which of the following statement is true.

- A** It is diatomic **B** it is nonmetal
C it is transition metal **D** it forms basic oxides

70 Geometrical isomerism in alkene is due to

- A** Unsaturated nature
C high reactivity
C Free rotation about C-C bond
D Restricted rotation of C=C bond

71 Which one of the following is general formula for ketone

- A** R_2O **B** $RCOH$
C R_2CO **D** RCO_2H
E all of these

72 EDTA is _____ ligand

- A** Bidentate **B** Tridentate
C Tetradentate **D** hexadentate

73 How many elements of periodic table have last electron of electronic configuration in s-orbital

- A** 12 **B** 10
C 13 **D** 14

74 The IUPAC name of given coordination compound is $[Co (NH_3)_3 (NO_2)_3]$

- A** triamine trinitro cobaltate (iii)
B triamine nitro cobalt (ii)
C triamine trinitrito cobalt (iii)
D trinitrito triamino cobalt (iii)

75 3d series elements are located at _____

- A** 3rd period **B** 4th period
C 2nd period **D** 5th period

76 Which of the following having highest binding energy

- A** Fe **B** Fe^{+2}
C Fe^{+3} **D** None of these

77 Which property of Zinc makes it able to be placed in transition metals

- A** Produce coloured compounds
B fixed oxidation Number
C Can make complex compounds
D it is typical element

78 Which of the following is incorrect about electro negativity?

- A** Metal has E-N value less than 2
C it has no unit
C With increase in O.S E.N decreases
D it determines polarity of molecule

79 Which of following group of elements have positive value of 1st electron affinity?

- A** IA **B** VIA
C IIA **D** VIIA

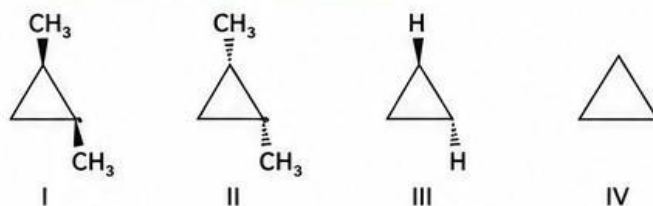
80 The element which has B.P less than $0^\circ C$ is

- A** Be **B** C
C B **D** N

81 Which of the following is not heterocyclic compound?

- A** Naphthalene **B** furan
C pyridine **D** pyrrole

82 Which of the following compound show geometric isomerism?



- A** Structure I **B** Structure II
C Structure III **D** Structure IV

83 Geminal and Vicinal dihalide are example related with type of isomerism

- A** Position isomerism
B Functional group isomerism
C metamerism
D Tautomerism
E not related with isomerism

84 First organic compound prepared in laboratory was

- A** urea **B** diamino methanone
C carbamide **D** all of these

85 The paramagnetic character of substance is due to the presence of

- A** bond pair of electrons
B lone pairs of electrons
C unpaired electron in the atom or molecule
D Pi electrons in atom.

86 Which pair of transition elements shows abnormal electronic configuration

- A** Sc and Zn **B** Zn and Cu
C Cu and Sc **D** Cu and Cr

- 87** The ability of transition element to form complexes is due to
- A** Small size metal ion
 - B** Highly charged ions
 - C** Availability of empty d-orbitals
 - D** all of these

- 88** which ion has all vacant d-orbital
- A** Sc^{+3}
 - B** Cu^{+1}
 - C** Zn^{+2}
 - D** Fe^{+2}

- 89** What is the correct order of electron affinity among O, F, Cl
- A** $\text{O} < \text{Cl} < \text{F}$
 - B** $\text{O} < \text{F} < \text{Cl}$
 - C** $\text{F} < \text{O} < \text{Cl}$
 - D** $\text{Cl} < \text{O} < \text{F}$

- 90** Which oxide is an amphoteric
- A** BaO
 - B** CaO
 - C** BeO
 - D** MgO

- 91** Beryllium shows the diagonal relationship with
- A** Na
 - B** Al
 - C** Mg
 - D** B

- 92** The element which resembles with both alkali metal and halogen is
- A** He
 - B** Cl
 - C** Na
 - D** H

- 93** The p-block elements along with s-block elements are collectively called as ____ elements
- A** Inner transition
 - B** Representative
 - C** Radioactive
 - D** transition

- 94** The outer shell electronic configuration d-block elements is given by ____
- A** $(n-1) d^{1-10} ns^2$
 - B** $(n-1) d^{1-10} (n-1) s^2$
 - C** $(n-1) d^{1-10} ns^{0-2}$
 - D** $nd^{1-10} ns^{0-2}$

- 95** Chalcogen are the elements of group ____
- A** 18
 - B** 16
 - C** 15
 - D** 12
 - E** 2

- 96** Which of the following magnetic property do alkaline earth metals exhibit
- A** paramagnetic
 - B** antiferromagnetic
 - C** diamagnetic
 - D** ferromagnetic

- 97** The two mirror images of molecule, which are non super imposable are called
- A** tautomerism
 - B** geometrical isomerism
 - C** enantiomers
 - D** isomers

- 98** Molecular formula which belongs with alkyne series is
- A** C_7H_{14}
 - B** $\text{C}_{10}\text{H}_{22}$
 - C** C_8H_{16}
 - D** $\text{C}_{25}\text{H}_{32}$

- 99** Pentane and 2-methyl butane have same
- A** Boiling point
 - B** Melting point
 - C** molecular formula
 - D** structural formula

- 100** The organic compounds having strongest functional group are
- A** carboxylic acid
 - B** ketone
 - C** aldehyde
 - D** alcohols

- 101** Which is true about transition elements
- A** these elements are inert in nature
 - B** these elements lose their electrons to become anions
 - C** they have variable oxidation states
 - D** they are highly reactive metals

- 102** An example of bidentate ligand among the following is ____
- A** EDTA
 - B** $\text{C}_2\text{O}_4^{2-}$
 - C** CN^-
 - D** H_2O

- 103** The highest oxidation state of chromium is
- A** +4
 - B** +5
 - C** +6
 - D** +7

- 104** Metamerism is not possible in
- A** Ether
 - B** ketone
 - C** carboxylic acid
 - D** amines

- 105** The number of unpaired electrons in Mn^{+2} is
- A** 2
 - B** 5
 - C** 3
 - D** 4

- 106** Two compounds having same composition, same molecular formula but different orientation in space are called
- A** Metamers
 - B** position isomer
 - C** structural isomers
 - D** stereoisomers

- 107** The isomers of substance have
- A** Same chemical properties
 - B** same molecular weight
 - C** Same structural formula
 - D** same functional groups

Which is incorrect statement about tautomers?

- $$\begin{array}{ccc} \text{O} & & \text{OH} \\ || & & | \\ \text{CH}_3-\text{C}-\text{CH}_2 & \rightleftharpoons & \text{CH}_3-\text{C}=\text{CH}_2 \\ \text{(Keto form)} & & \text{(Enol form)} \end{array}$$

109 In which sets of ions paramagnetic property increases?

- A** $\text{Cu}^{+2} < \text{V}^{+3} < \text{Cr}^{+2} < \text{Mn}^{+2}$
- B** $\text{Cu}^{+2} < \text{Cr}^{+2} < \text{V}^{+3} < \text{Mn}^{+2}$
- C** $\text{Mn}^{+2} < \text{V}^{+3} < \text{Cr}^{+2} < \text{Cu}^{+2}$
- D** $\text{Mn}^{+2} < \text{Cu}^{+2} < \text{Cr}^{+2} < \text{V}^{+3}$

110 Which one of the following ions has electronic configuration $3d^5$?

- A** Co^{+3} **B** Ni^{+3}
C Mn^{+3} **D** Fe^{+3}

111 Which of the following compounds does not react with H_2SO_4

- A** Alkanes **B** alcohol
C alkenes **D** alkyne

112 The IUPAC name of $\text{K}_2 [\text{Ni} (\text{CN})_4]$ is _____

- A** Potassium tetracyanonickelate (III)
B Potassium tetracyanonickelate (II)
C Potassium tetracyanonickelate (III)
D Potassium tetracyanonickel (II)

113 The chlorination of methane is

- A** An electrophilic addition
B A free radical substitution
C A nucleophilic addition
D An electrophilic substitution
- $$\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{h}\nu} \text{CH}_3\text{Cl} + \text{HCl}$$
- (Methane) (Chloromethane)

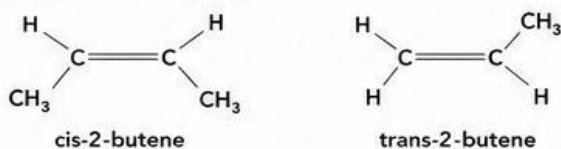
114 The combustion of pentane produces

- A** Pentene **B** $\text{HCl} + \text{H}_2\text{O}$
C Pentyne **D** $\text{CO}_2 + \text{H}_2\text{O}$

115 Octane Number scale is used to check quality of

- A** Gasoline **B** kerosene oil
C diesel oil **D** lubricating oil

116 How do the M.P of trans isomers compare to the cis isomer for alkene.



- A** cis isomers have high M.P
B trans isomers have high M.P
C Both have similar M.P
D none of these

117 Super oxide is produced by

- A** Be **B** Fr
C Ba **D** K

118 Which of the following is most electropositive properties?

- (A)** Cs **(B)** Fr
(C) Ba **(D)** Na

119 Which elements of group II A produce normal oxide with oxygen?

- A** Li **B** Sr
C Ba **D** Be

120 Which element produces non stoichiometric compound with transition elements

- A** H **B** B
C C **D** N
E All

121 Which is general formula when nitrogen reacts with group II A

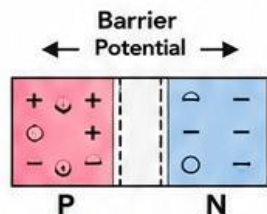
- A** MN_3
B M_3N_2
- C** M_2N_2
D M_3N_2

122 Transition metals in complex Compounds act as

- A** Lewis acid **B** lewis base
C neutral **D** none of these

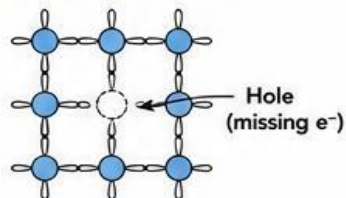
123 In a semiconductor diode, the barrier potential offers opposition to

- A Holes in P-region only
- B free electrons in N-region only
- C Majority carriers in both regions
- D majority as well as minority carriers in both regions



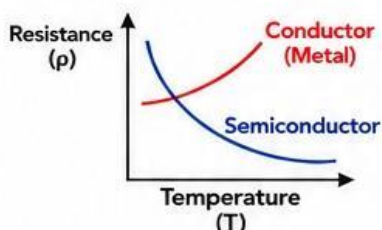
124 In a junction diode, the holes are due to

- A Protons
- B extra electrons
- C neutrons
- D missing electrons



125 By increasing the temperature, the specific resistance of a conductor and a semiconductor

- A Increases for both
- B decreases for both
- C Increases, decreases
- D decreases, increases



126 In the half wave rectifier circuit operating from 50 Hz mains frequency, the fundamental frequency in the ripple would be

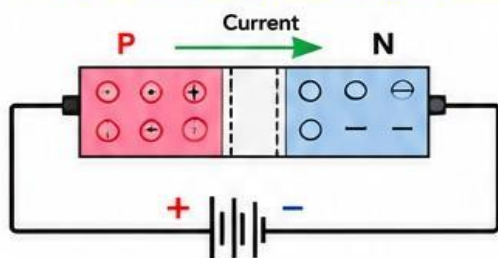
- A 25 Hz
- B 50 Hz
- C 70.7 Hz
- D 100 Hz

127 In a full wave rectifier circuit operating from 50 Hz mains frequency, the fundamental frequency in the ripple would be

- A 25 Hz
- B 50 Hz
- C 70.7 Hz
- D 100 Hz

128 In forward biasing of the p-n junction

- A The positive terminal of the battery is connected to p-side and the depletion region becomes thick
- B The positive terminal of the battery is connected to n-side and the depletion region becomes thin
- C The positive terminal of the battery is connected to n-side and the depletion region becomes thick
- D The positive terminal of the battery is connected to p-side and the depletion region becomes thin



129 If a small amount of antimony is added to germanium crystal

- A It becomes a p-type semiconductor
- B The antimony becomes an acceptor atom
- C There will be more free electrons than holes in the semiconductor
- D Its resistance is increased

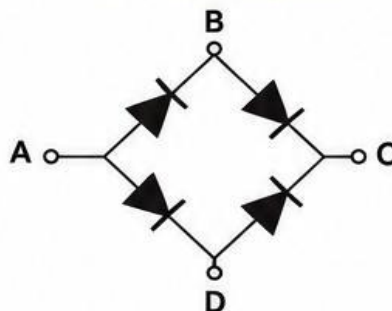
130 At absolute zero, Si acts as

- A non-metal
- B metal
- C insulator
- D none of these

131 The difference in the variation of resistance with temperature in a metal and a semiconductor arises essentially due to the difference in the

- A crystal structure
- B variation of the number of charge carriers with temperature
- C type of bonding
- D variation of scattering mechanism with temperature

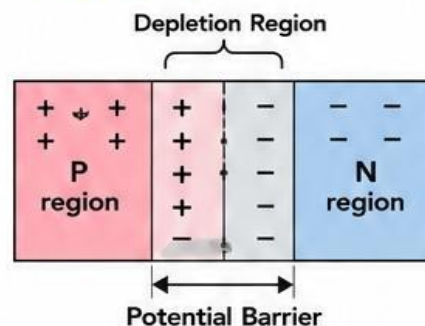
132 In the diagram, the input is across the terminals A and C and the output is across B and D. Then the output is



- A zero
- B same as the input
- C full wave rectifier
- D half wave rectifier

133 The cause of the potential barrier in a p-n diode is

- A depletion of positive charges near the junction
- B concentration of positive charges near the junction
- C depletion of negative charges near the junction
- D concentration of positive and negative charges near the junction



134 Application of a forward bias to a p-n junction

- A** widens the depletion zone
- B** increases the potential difference across the depletion zone
- C** increases the number of donors on the n side
- D** increases the electric field in the depletion zone.

135 A semi-conducting device is connected in a series circuit with a battery and a resistance. A current is found to pass through the circuit. If the polarity of the battery is reversed, the current drops to almost zero. The device may be

- A** a p-n junction
- B** an intrinsic semi-conductor
- C** a p-type semi-conductor
- D** an n-type semi-conductor

136 A solid that is not transparent to visible light and whose electrical conductivity increases with temperature is formed by

- A** ionic binding
- B** covalent binding
- C** metallic binding
- D** vander Waal's binding

137 In a P -N junction

- A** the potential of P & N sides becomes higher alternately
- B** the P side is at higher electrical potential than N side.
- C** the N side is at higher electrical potential than P side.
- D** both P & N sides are at same potential

138 Barrier potential of a P-N junction diode does not depend on

- A** doping density
- B** diode design
- C** temperature
- D** forward bias

139 Reverse bias applied to a junction diode

- A** increases the minority carrier current
- B** lowers the potential barrier
- C** raises the potential barrier
- D** increases the majority carrier current

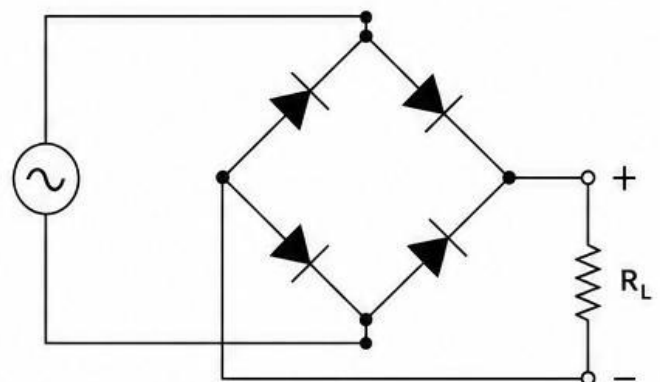
140 Which one of the following statement is false?

- A** Pure Si doped with trivalent impurities gives a p-type semiconductor
- B** Majority carriers in a n-type semiconductor are holes
- C** Minority carriers in a p-type semiconductor are electrons
- D** The resistance of intrinsic semiconductor decreases with increase of temperature

141 In a n-type semiconductor, which of the following statement is true?

- A** Electrons are minority carriers and pentavalent atoms are dopants.
- B** Holes are minority carriers and pentavalent atoms are dopants.
- C** Holes are majority carriers and trivalent atoms are dopants.
- D** Electrons are majority carriers and trivalent atoms are dopants.

142 In a bridge rectifier, the number of diodes required is



- A** 1
- B** 2
- C** 3
- D** 4

143 Directions for question contains STATEMENT-1 and STATEMENT-2.

Statement-1: When two semiconductor of p and n type are brought in contact, they form p-n junction which act like a rectifier.

Statement-2: A rectifier is used to convert alternating current into direct current.

Choose the correct answer (ONLY ONE option is correct) from the following-

- A Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1
- B Statement -1 is true, Statement-2 is true; Statement -2 is not a correct explanation for Statement-1
- C Statement -1 is true, Statement-2 is false
- D Statement -1 is false, Statement-2 is true

144 Directions for Qs each question contains STATEMENT-1 and STATEMENT-2.

Statement 1: A pure semiconductor has negative temperature coefficient of resistance.

Statement 2: In a semiconductor on raising the temperature, more charge carriers are released, conductance increases and resistance decreases.

Choose the correct answer (ONLY ONE option is correct) from the following-

- A Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1
- B Statement -1 is true, Statement-2 is true; Statement -2 is not a correct explanation for Statement-1
- C Statement -1 is true, Statement-2 is false
- D Statement -1 is false, Statement-2 is true

145 Directions for Qs each question contains STATEMENT-1 and STATEMENT-2.

Statement 1: If the temperature of a semiconductor is increased then its resistance decreases.

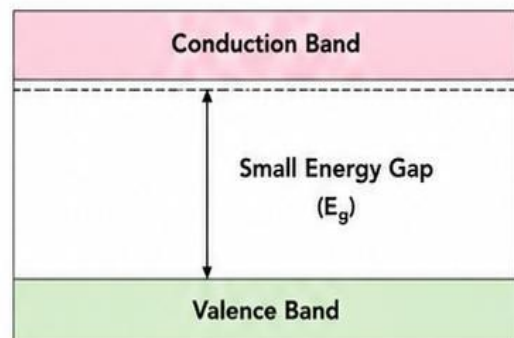
Statement 2: The energy gap between conduction band and valence band is very small.

Choose the correct answer (ONLY ONE option is correct) from the following-

- A Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1
- B Statement -1 is true, Statement-2 is true; Statement -2 is not a correct explanation for Statement-1
- C Statement -1 is true, Statement-2 is false
- D Statement -1 is false, Statement-2 is true

146 Hole in semiconductor is

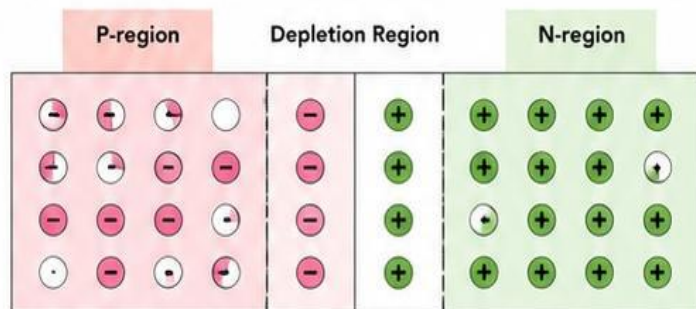
- A an anti - particle of electron
- B a vacancy created when an electron leaves a covalent bond



- C absence of free electrons
- D an artificially created particle

147 In an unbiased p-n junction, holes diffuse from the p-region to n-region because of

- A the potential difference across the p-n junction
- B the attraction of free electrons of n-region
- C the higher hole concentration in p-region than that in n-region
- D the higher concentration of electrons in the n-region than that in the p-region



Red circle with minus sign: Ionized Acceptor

White circle: Hole

Green circle with plus sign: Ionized Donor

White circle: Electron

148 The barrier potential of a p-n junction depends on:

- a) type of semiconductor material
- b) amount of doping
- c) temperature

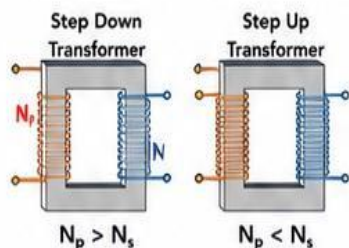
Which one of the following is correct ?

- A a and b only
- B b only
- C b and c only
- D a, b and c

149 Which of the following statement is false?

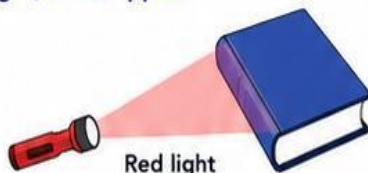
For step down transformer ----- for step up transformer

- A** $E_p/E_s > 1$ ----- $E_p/E_s < 1$
B $N_s/N_p > 1$ ----- $N_s/N_p < 1$
C $I_s/I_p > 1$ ----- $I_s/I_p < 1$
D None of these



150 If blue book is seen with red light, it will appear:

- A** blue
B red
C black
D white



151 A hydrogen atom in its ground state absorbs 10.2 eV of energy. What is the orbital angular momentum is increased by?

- A** 4.22×10^{-34} Js **B** 2.11×10^{-34} Js
C 3.16×10^{-34} Js **D** 1.05×10^{-34} Js

152 The equation of Rydberg constant is given by:

- A** $R_H = \frac{h_e}{m_e}$ **B** $R_H = \frac{E_e}{hc}$
C $R_H = \frac{E_e}{\lambda}$ **D** $R_H = \frac{1}{hc}$

153 Find out the minimum energy required to take out the electron from first excited state of He^+ :

- A** 13.6 eV **B** 122.4 eV
C 54.4 eV **D** 3.4 eV

154 According to Bohr's principle, what is the relation between the principal quantum number "n" and the linear momentum "P" of electron in nth orbit?

- A** P proportional to $1/n$ **B** P proportional to $1/n^2$
C P proportional to n **D** P proportional to n^2

155 The total number of spectral lines for an electron transition from $n=5$ to $n=3$ states is:

- A** 3 **B** 5 **C** 8 **D** 10

156 What is the energy required to ionize an H-atom from the third excited state, if ground state ionization energy of H-atom is 13.6 eV?

- A** 1.5 eV **B** 0.85 eV
C 13.6 eV **D** 3.4 eV

157 Paschen series is obtained when all the transition of electron terminate on:

- A** 2nd orbit **B** 3rd orbit
C 4th orbit **D** 5th orbit

158 Speed of electron in first Bohr's orbit is:

- A** 2.2×10^6 ms⁻¹ **B** 2.2×10^{10} ms⁻¹
C 2.2×10^8 cms⁻¹ **D** 2.2×10^6 cms⁻¹

159 Find the true statement.

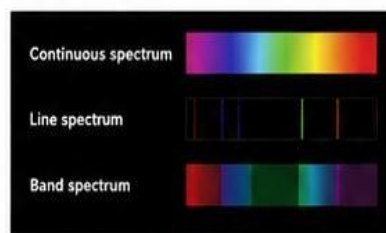
- A** An electron will not lose energy when jumping from the 1st orbit to the 3rd orbit
B An electron will not give energy when jumping from the 1st orbit to the 3rd orbit
C An electron will release energy when jumping from the 1st orbit to the 3rd orbit
D An electron will absorb energy when jumping from the 1st orbit to the 3rd orbit

160 A photon of maximum frequency is emitted when electron in Hydrogen atom jumps from orbit:

- A** $n = 1$ to $n = 3$
B $n = 2$ to $n = 6$
C $n = 7$ to $n = 1$
D $n = \infty$ to $n = 3$

161 Atomic spectra are the examples of:

- A** Continuous spectra
B Line spectra
C Band spectra
D Mix spectra

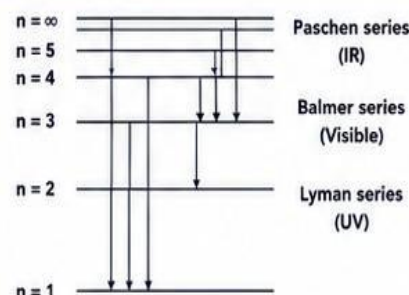


162 If electron jumps from second orbit to first orbit in hydrogen atom it emits photon of:

- A** 3.4 eV **B** 10.20 eV
C 13.6 eV **D** 12.09 eV

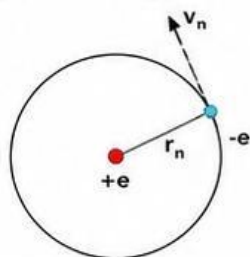
163 In which region Balmer series lies:

- A** Visible
B far infrared
C Ultraviolet
D Infrared



- 164** The orbital angular momentum in the allowed stationary orbits of hydrogen atom is given as:

- A $2\pi/nh$
 B $nh/2\pi$
 C $\pi/2h$
 D π/h



- 165** The radius of 10th orbit in hydrogen atom is:

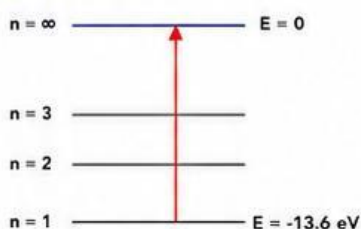
- A 0.053m
 B 0.053nm
 C 5.3nm
 D 53nm

$$r_n = a_0 n^2$$

$$a_0 = 0.053 \text{ nm}$$

- 166** The potential required to remove an electron from the atom completely is called:

- A critical potential
 B Ionization potential
 C absolute Potential
 D Excitation Potential

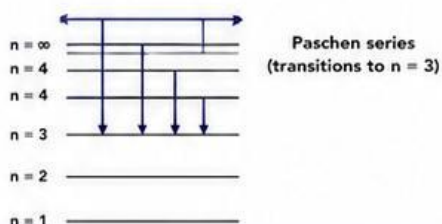


- 167** In an electronic transition, a atom cannot emit:

- A Y-rays
 B Infrared rays
 C UV-rays
 D X-rays

- 168** The maximum wavelength of Paschen series is:

- A $\frac{9}{7R}$
 B $\frac{144}{7R}$
 C $\frac{144}{R}$
 D $\frac{9}{R}$



- 169** Which of the following is/are correct for n^{th} orbit of Hydrogen atom

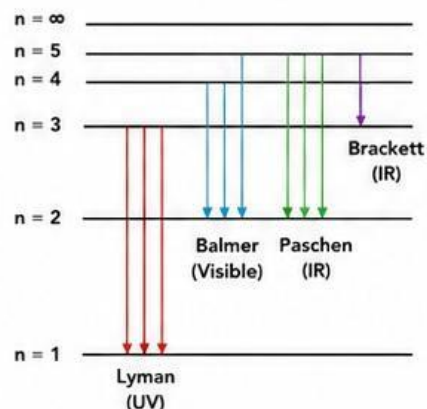
- I) Kinetic energy is equal to total energy and is taken as positive
 II) Kinetic energy is equal to twice of total energy and is taken as positive
 III) Potential energy is equal to twice of total energy and is taken as negative
 IV) Potential energy is equal to total energy and is taken as negative
- A I only
 B IV only
 C I and III only
 D II and IV only

- 170** First spectral series of hydrogen atom was discovered by:

- A Balmer
 B Lyman
 C Paschen
 D Rydberg

- 171** Which of the following spectral series lies in the ultraviolet region:

- A Pfund series
 B Paschen series
 C Balmer series
 D Lyman series

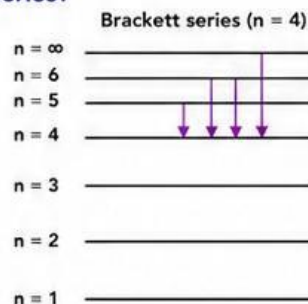


- 172** The total energy of electron in the state $n = \text{infinity}$ of the hydrogen atom is:

- A zero
 B 3.2 eV
 C 10.2 eV
 D 13.6 eV

- 173** What is the ratio of minimum to maximum wavelength in the Brackett Series?

- A 25 : 9
 B 5 : 36
 C 9 : 25
 D 3 : 4



- 174** When a hydrogen atom is in its first excited level, what is the relation of radius of that state level with Bohr's radius?

- A Twice
 B 4 times
 C Same
 D Half

$$r_n = a_0 n^2$$

First excited level $\Rightarrow n = 2$

$$r_2 = 4a_0$$

- 175** Which of the following is true regarding the Bohr model of atoms?

- A Assumes that the angular momentum of electrons is quantized
 B Uses Faraday's laws
 C Predicts continuous emission spectra for atoms
 D Predicts the same emission spectra for all types of atoms

- 176** In which of the following system, will the radius of the first orbit ($n = 1$) be minimum?

- A Doubly-ionized Lithium (Li^{2+})
 B Singly-ionized helium (He^+)
 C Deuterium atom (^2H)
 D Hydrogen atom (H)

$$r_1 = \frac{a_0}{Z}$$

$$\text{Radius} \propto \frac{1}{Z}$$

Larger $Z \Rightarrow$ smaller r_1

177 Choose the most appropriate statement.

- A I know her for years.
- B I knew her for years.
- C I have known her for years.
- D I am knowing her for years.

178 Mozart *_*_ more than 600 pieces of music.

- A wrote
- B has written
- C had written
- D NOT

179 He hurt his back while he was working in the garden.

- A hurt
- B while
- C was working
- D No error

180 Choose the most appropriate answer.

- A She is studying since yesterday.
- B She has been studying since yesterday
- C She studied since yesterday
- D All of the above

181 Find the appropriate statement:

- A By the time you read this, I will be on the plane.
- B Until you will read this, I will be on the plane.
- C As soon as you read this, I am on the plane.
- D If you read this, I was on the plane

182 He has been working out a lot; he *_*_ really fit.

- A Will get
- B is going to get
- C gets
- D got

183 She walks as though she is an actress.

- A walks
- B though
- C is
- D an
- E No error

184 I promise I will finish the report by tomorrow.

- A Promise
- B will finish
- C by
- D No error

185 He didn't know that honesty was the best policy.

- A Didn't
- B that
- C was
- D No error

186 Choose the appropriate statement.

- A The cost of living is increasing nowadays.
- B The cost of living was increasing nowadays.
- C The cost of living has been increasing nowadays.
- D The cost of living increases nowadays.

187 I have been seeing a lot of Sajid since he moved into the flat upstairs.

- A have
- B seeing
- C since
- D moved into
- E No error

188 People have been saying for ages that the building should be pulled down.

- A Have been
- B for ages
- C should be
- D pulled
- E No error

189 When I was a child we were walking to school every day.

- A When
- B was
- C were walking
- D every day
- E No error

190 I _____ "Avengers" movie six times.

- A Watch
- B watched
- C have watched
- D am watching

191 I will read this book after I will have dinner.

- A Will
- B after
- C will
- D NO error

192 Find the most appropriate statement.

- A They have been friends in 2020.
- B They were friends since 2020.
- C They used to be friends for years.
- D They are friends since childhood.

193 She is used to woke up in the middle of the night.

- A is
- B used to
- C woke
- D in the middle
- E No error

194 I _____ of living in Ireland even before I _____ the country.

- A had dreamed/ visited
- B dreamed/ had visited
- C dreamed/ visited
- D dream/ visited

— PASSAGE —

Aria and her family are gathered around the television. They are watching a movie about a tornado. A tornado is a big funnel of wind. Tornadoes travel over the ground and can move very fast. They can be very destructive, and they damage almost everything they touch.

In the movie Aria and her family are watching, a big tornado comes through a small town. The tornado uproots trees, destroys houses, and harms animals. It even drains a lake! (This part was a little hard to believe.) Anyway, the tornado causes lots of destruction. It is pretty scary, and no one can stop it. It just keeps going and going.

A little girl lives in the small town. The tornado hasn't come near her yet, but it is on its way. The little girl and her family are in danger. The tornado looks like it is going to hit their house. The girl is scared, but she knows she can't let this happen. She must do something to **prevent** it. She tries to think, but she doesn't know how to stop it.

Then something amazing happens. The little girl builds a kite using tent poles and the sheets from her bed. The kite she makes is giant – it is bigger than her house! She flies the kite into the tornado. The kite is so big that it steals all the wind from the tornado. The tornado dies, and the little girl saves the town.

Then the movie ends. "Time to brush your teeth," says Aria's dad. "Then it's off to bed, kiddo."

"Dad," Aria says. She is still thinking about the movie. "Can a kite really stop a tornado?" "No," he says. "But little girls really can do amazing things."

195 In this passage, the author mostly

- A defines a tornado
- B summarizes a movie
- C describes a girl
- D answers a question

196 As used in paragraph 4, the word **prevent** most nearly means

- A stop
- B delay
- C allow
- D help

197 What category does the movie described in this passage belong to?

- A mystery
- B romance
- C action
- D comedy

198 Which detail about the movie is probably NOT true?

- A The tornado uproots trees.
- B The tornado destroys houses.
- C The tornado harms animals.
- D The tornado drains a lake.

199 Statement:

Fact 1: Maria said "Anila and I both have cats".

Fact 2: Anila said "I don't have a cat"

Fact 3: Maria always tells the truth, but Anila sometimes.

If the first three statements are facts, which of the following must be a fact?

- 1. Anila has a cat
- 2. Maria has a cat
- 3. Anila is lying

- A I only
- B II only
- C I and II only
- D All statements are facts

200 Statement:

Battery X lasts longer than Y.

Battery Y doesn't last as long as battery Z

Battery Z lasts longer than X.

If the first two are true, the third statement is.

- A True
- B false
- C uncertain
- D none of these



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MOCK TEST 4

ANSWER KEY



Q.	ANS	Q.	ANS	Q.	ANS	Q.	ANS	Q.	ANS	Q.	ANS	Q.	ANS	Q.	ANS
1	D	26	A	51	A	76	C	101	C	126	B	151	D	176	A
2	D	27	C	52	A	77	C	102	B	127	D	152	B	177	C
3	B	28	D	53	A	78	C	103	C	128	D	153	A	178	A
4	A	29	D	54	B	79	C	104	C	129	C	154	A	179	D
5	B	30	A	55	C	80	D	105	D	130	C	155	A	180	B
6	B	31	A	56	C	81	A	106	D	131	B	156	B	181	A
7	A	32	A	57	D	82	A	107	B	132	C	157	B	182	B
8	C	33	D	58	D	83	A	108	D	133	D	158	A	183	C
9	D	34	A	59	C	84	D	109	A	134	C	159	D	184	D
10	A	35	A	60	C	85	C	110	A	135	A	160	C	185	C
11	A	36	B	61	D	86	D	111	A	136	B	161	B	186	A
12	B	37	A	62	A	87	C	112	B	137	B	162	D	187	E
13	D	38	D	63	C	88	A	113	B	138	B	163	A	188	E
14	D	39	D	64	C	89	B	114	D	139	C	164	B	189	C
15	A	40	B	65	A	90	C	115	A	140	B	165	C	190	C
16	C	41	B	66	D	91	B	116	A	141	B	166	B	191	C
17	A	42	C	67	A	92	D	117	D	142	D	167	A	192	C
18	D	43	D	68	A	93	B	118	A	143	B	168	C	193	B
19	D	44	A	69	D	94	C	119	D	144	A	169	C	194	A
20	D	45	A	70	D	95	B	120	E	145	A	170	A	195	B
21	E	46	B	71	C	96	C	121	C	146	B	171	D	196	A
22	A	47	D	72	D	97	C	122	A	147	C	172	A	197	C
23	A	48	B	73	D	98	C	123	C	148	D	173	C	198	D
24	B	49	Cancelled	74	C	99	C	124	D	149	B	174	B	199	D
25	D	50	C	75	B	100	A	125	C	150	C	175	A	200	C



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